Paul He

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Education

Nanyang Technological University, PhD in Computer Science — Jan 2026 – Dec 2030
Supervisor: Prof. Kwok Yan Lam (College of Computing and Data Science) — Singapore

- Incoming PhD student with full research scholarship

University of Toronto, BSc in Computer Science (Honours) — Sep 2021 – May 2025
Supervisors: Prof. Gerald Penn, Prof. Zhijing Jin — Toronto, Canada

- Graduated with High Distinction

ETH Zürich, Exchange Year in Computer Science — Sep 2023 – Aug 2024

- Completed graduate-level coursework in NLP and Machine Learning

Work Experience

Amazon Web Services (AWS) — Sep 2025 – Dec 2025
Applied Scientist II, CloudWatch Logs — Tübingen, Germany

- Conducted open-ended research on reliability and evaluation of large language models in production settings.
- Implemented novel evaluation and reliability methods as production-quality code in internal pipelines.
- Contributed research directions that developed into ongoing manuscripts and collaborations.

Research Experience

Causal Reasoning Evaluation for Language Models — Jan 2025 – Aug 2025
Supervisor: Prof. Zhijing Jin (Jinesis AI Lab) — Vector Institute

- Designed a sound & complete symbolic verification metric to uncover correctness missed by standard evaluations.

Polynomial-Time CCG Parsing — Sep 2024 – Aug 2025
Supervisor: Prof. Gerald Penn — University of Toronto

- Conducted large-scale multilingual experiments identifying empirical complexity thresholds where symbolic parsing outperforms naive baselines.

Unsupervised Prompt Optimization for RAG using PMI — Jun 2024 – Sep 2024
Supervisors: Tianyu Liu, Prof. Ryan Cotterell (Rycolab) — ETH Zürich

- Developed an unsupervised PMI-based document reordering method, improving QA accuracy by up to 4% on NQ-Open and ELI5.

Publications

Uncovering Hidden Correctness in LLM Causal Reasoning via Symbolic Verification
Paul He, Yinya Huang, Mrinmaya Sachan, Zhijing Jin. *NeurIPS 2025 CauScien (non-archival); under review at ARR.*

CCG Revisited: A Multilingual Empirical Study of the Kuhlmann–Satta Algorithm
Paul He, Gerald Penn. *IWPT 2025.* aclanthology.org/2025.iwpt-1.3

Pointwise Mutual Information as a Performance Gauge for Retrieval-Augmented Generation
Tianyu Liu*, Jirui Qi*, **Paul He**, Arianna Bisazza, Mrinmaya Sachan, Ryan Cotterell. *NAACL 2025 (Main).* aclanthology.org/2025.naacl-long.78

Preprints & Under Review

Dialogue as Uncertainty Reduction: A Judge-Free Metric for Multi-Turn Dialogue Evaluation
Paul He, Shiva Kasiviswanathan‡, Dominik Janzing‡. *Manuscript in preparation for submission to ARR (Jan 2026).*

Foundations of Global Consistency Checking with Noisy LLM Oracles
Paul He, Elke Kirschbaum, Shiva Kasiviswanathan. *Under review at ARR.*

** Equal contribution. ‡ Equal supervision.*

Skills

Programming Languages: Python (research & production), C, Java, SQL

Languages: English (fluent), Mandarin (proficient), German (proficient)